Smart Parking Availability Prediction Using Machine Learning

Course: ML 402331 – IoT Machine Learning

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Project Track: Software-only

Project Idea

This project aims to develop a system that uses machine learning to predict how full a parking area will be. It will use previous parking records from Birmingham, UK, to estimate the number of occupied spaces at a certain time.

The system will use the predicted number of occupied spaces and the total capacity of the car park to estimate how many spaces are still available.

Finding an available parking space can sometimes be difficult, especially during busy hours. Drivers may waste time travelling to a car park without knowing whether it has available spaces. This project aims to help drivers make better decisions by giving them an estimate of parking availability before they arrive.

The project will be software-based and will not require physical sensors or hardware.

Problem Statement and Research Question

Problem Statement: Traditional parking monitoring requires manual checks or costlyinfrastructure. While automated ultrasonic sensors are affordable, their data can be noisy orprone to environmental errors. A reliable machine learning pipeline is needed to clean thisdata, classify parking occupancy accurately, and catch anomalies.

Research Question: How effectively can machine learning classifiers predict parking slotoccupancy from sensor data, and can clustering methods reliably flag unusual sensorreadings or faults?

Machine Learning Task and Features

The project will use supervised regression. This means the model will learn from previous records where the actual parking occupancy is already known. It will then use what it has learned to predict occupancy for other records.

The main variables will be:

- Target: Occupancy – the number of spaces currently being used.

- SystemCodeNumber: Identifies the car park.

- Capacity: The total number of spaces in the car park.

- Hour: The hour when the parking record was collected.

- Minute: The minute of the recorded time.

- DayOfWeek: The day of the week when the record was collected.

The original dataset contains four columns: SystemCodeNumber, Capacity, *Occupancy* and LastUpdated. The time-related features will be created from LastUpdated.

(Occupancy will be the value the model tries to predict, so it will not be used as an input feature)

Dataset and Data Plan

The project will use the Parking Birmingham dataset from the UCI Machine Learning Repository. It is freely available and contains approximately 35,717 parking records from car parks in Birmingham, UK.

The dataset includes the car park name, its total capacity, the number of occupied spaces and the time of each record.

The dataset is suitable for this project because it already contains the information needed to study changes in parking occupancy over time.

The data will be checked for missing or incorrect values before being prepared for the machine learning model. The time column will also be used to create additional features.

(In a later phase, Python will be used to replay the historical records one by one to imitate new parking updates. This will be a simulation and will not use information from real-time parking sensors)

Dataset link:

https://archive.ics.uci.edu/dataset/482/parking+birmingham

Planned Work

The first model will be Linear Regression. In the later phases, other models such as Ridge, Lasso, Random Forest and Gradient Boosting will be tested. Their results will be compared to find which model gives better predictions.

Related Work

1. Stolfi, D. H., Alba, E., & Yao, X. (2017). "Predicting Car Park Occupancy Rates in Smart Cities." Smart Cities: Second International Conference (Smart-CT 2017), 107–117.

https://doi.org/10.1007/978-3-319-59513-9_11

This research focuses on predicting how full car parks are. It is related to this project because it studies the same general problem of estimating parking occupancy.

2. Koumetio Tekouabou, S. C., Abdellaoui Alaoui, E. A., Cherif, W., & Silkan, H. (2022). "Improving parking availability prediction in smart cities with IoT and ensemble-based model." Journal of King Saud University – Computer and Information Sciences, 34(3), 687–697.

https://doi.org/10.1016/j.jksuci.2020.01.008

3. Dahiya, A., et al. (2024). "Machine Learning-Based Prediction of Parking Space Availability in IoT-Enabled Smart Parking Management Systems." Journal of Advanced Transportation, 2024, Article 8474973.

https://doi.org/10.1155/2024/8474973

Expected Results and Limitations

The expected result is a simple software system that can estimate the number of occupied and available parking spaces using previous parking records.

The project will start with a basic Linear Regression model and later test other models to compare their results.

One limitation is that the dataset contains historical records from Birmingham only. Therefore, the predictions may not represent parking conditions in other cities or current parking situations. The project will demonstrate how machine learning can be used for parking predictions rather than provide a live parking service.

AI Assistance

AI tools were used to help organise and simplify parts of the proposal.